

The Writer — first inside, alone



WRITER

Enters the GPS-denied space with no map and no operator. Explores, senses, decides what matters, and leaves its knowledge behind.

1

Explore autonomously

navigate an unknown, GPS-denied space alone

2

Sense events

detect at least 2 event types (hazard, victim, hole, gas, junction, exit...)

3

Decide what matters

choose in real time which findings are worth recording

4

Drop beacons

deposit a beacon where each event is, then return to the entrance



A few bytes that describe themselves

Each beacon is a tiny ground transmitter that just broadcasts a compact message.
The message says what, where and when.

A COMPACT BEACON MESSAGE

0	identifier	which beacon is speaking
1	WHAT	hazard · junction · victim · exit...
2	WHERE	direction to take from here
3	distance	metres to next beacon / target
4	WHEN	timestamp this is what makes it age
5	version	lets a later agent overwrite it

The SAME beacon, at different ages:

FRESH · 2 min

"Hazard ahead — go right, 4 m." Trust as measured.

AGING · 15 min

"Possible hazard — verify before crossing."

STALE · 40 min

"Old warning — rely on your own senses."

SUSPECT · contradicted

Live sensor disagrees flag it, discard.

The Executor — enters with a mission

1

Receives a mission

extinguish • extract • seal • retrieve — tied to the chosen scenario

2

Enters informed

no blind search: it already knows what to reach and what to avoid

3

Navigates by beacons

the beacons are the map it listens and follows, no GPS needed

4

Trusts by age

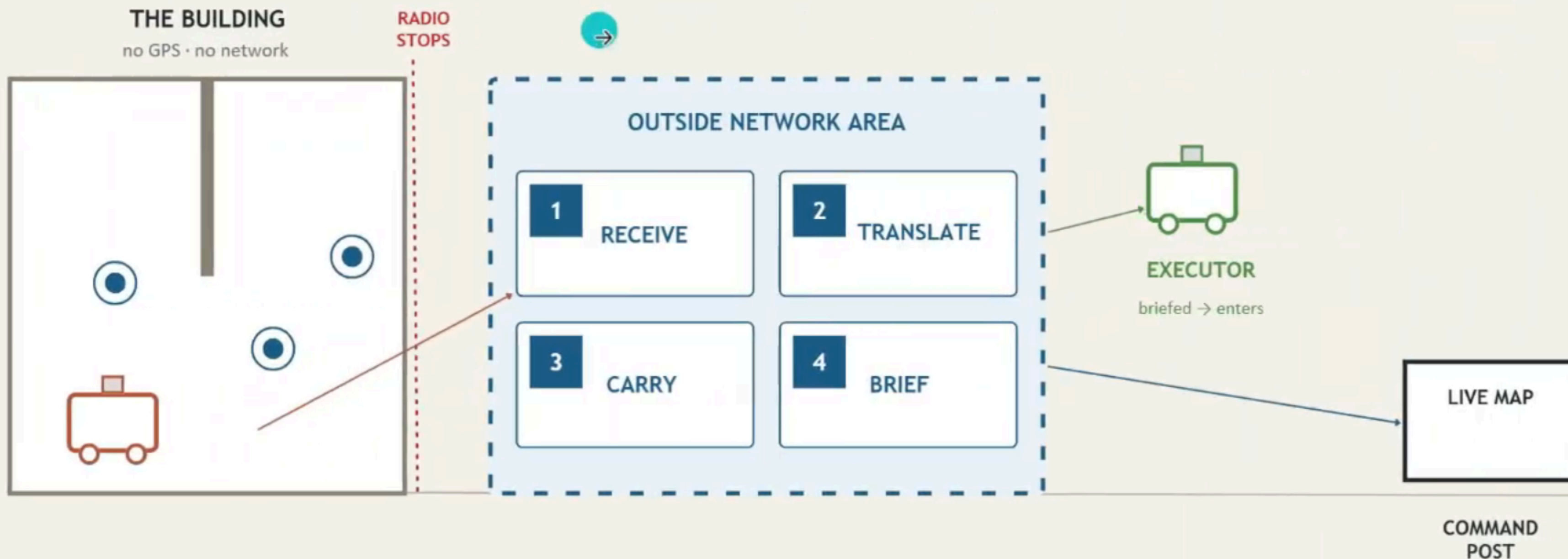
fresh beacons guide it; stale ones it verifies with its own sensors



EXECUTOR

Briefed by the command post before entry, it walks in already knowing the plan then follows the beacons to act.

The system chain



Writer explores → beacons deposited → outside area receives & translates → command post → Executor navigates by beacons

The Outside Network Area

Inside: no GPS, no network. Outside: GPS, satellite time and communication are available. This area is the bridge that connects the two and teams design its interior. It is defined by its ROLES, not its parts. →

1 RECEIVE

Take in what the Writer produced when it returns — events, beacons, its private frame.

2 TRANSLATE

Convert the robot's private coordinates into a real-world GPS reference anyone can use.

3 CARRY

Transfer the information to a distant command post — wireless or satellite links.

4 BRIEF

Prepare and deliver the mission plan the Executor needs before it enters the space.

All communication between inside and outside passes through this area — no direct link between the robots and the command post.

What each team must do

- ✓ Choose ONE environment
- ✓ Detect at least 2 event types
- ✓ Use at least 2 physical robots Writer & Executor
- ✓ Implement a functional Outside Network Area
- ✓ Route all inside↔outside communication through that area
- ✓ No direct link between the robots and the command post
- ✓ Prepare and demonstrate failure cases

ENVIRONMENTS

Urban Search & Rescue

Fire / Hazardous Building

Mines / Tunnels

Nuclear / Contaminated

TECHNICAL FOCUS

The areas your solution must cover

Writer Robot

autonomous navigation · exploration · decision-making in GPS-denied environments

Event Detection & Beacon Deposition

sensing · deciding what to record · beacon drop mechanism

Beacon Design

compact message · RF broadcast · message aging mechanism

Frame Translation

private robot coordinates → real-world GPS coordinates

Outside Network Area

command-post connection · satellite / wireless · data reception & forwarding

Executor Robot

mission briefing · beacon-guided navigation · mission execution

Technical Documentation

system architecture · data flow · additional technical diagrams

What you submit first

REQUIRED DELIVERABLES

- GitHub repository
- Simulation demo
- Short technical report max. 6 pages
- Implementation plan
- Failure cases
- Technical solution & architecture

The solution must address:

- › Writer robot autonomy
- › Event detection & beacon deposition
- › Beacon message & signal design
- › Frame translation
- › Outside network architecture
- › Executor robot
- › System architecture & data flow